

Week 2 Report – Data Collection Pipelines

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Week: 2

Topics Covered

- ✧ Web Scraping
- ✧ JSON APIs
- ✧ Data Ingestion
- ✧ Handling Pagination

Libraries Used

- ✧ requests
- ✧ BeautifulSoup
- ✧ pandas

Work Completed

During Week 2, I learned how to collect data from both APIs and websites. I used the `requests` library to fetch data, `BeautifulSoup` to scrape HTML content, and `pandas` to organize the collected data into a structured DataFrame and export it as a CSV file.

I practiced working with JSON APIs by retrieving data, converting it into a DataFrame, and saving it to a CSV file.

For web scraping, I built a crawler using the Quotes to Scrape website. The crawler visited multiple pages using pagination, extracted quotes and authors, stored the data in a list of dictionaries, converted it into a DataFrame, and exported the final dataset to a CSV file.

I also implemented duplicate handling using `drop_duplicates()` and handled missing text safely by checking whether an element existed before accessing its text. If a quote or author was missing, the program stored "Not Found" instead of crashing.

Assignment Completed

- ✧ Built a web crawler
- ✧ Implemented pagination
- ✧ Collected and stored scraped data

- ✧ Exported dataset to CSV
- ✧ Handled duplicate records
- ✧ Handled missing text

Skills Gained

- ✧ API data collection
- ✧ Web scraping with BeautifulSoup
- ✧ Data ingestion using pandas
- ✧ Pagination handling
- ✧ Duplicate data removal
- ✧ Missing data handling
- ✧ CSV dataset creation

Code Implementation Screenshots

```
crawler.py M X
Week_2 > crawler.py > ...
16
17 import requests
18 from bs4 import BeautifulSoup
19 import pandas as pd
20
21 data = []
22
23 for page in range(1, 11):
24     # HTML Website
25     url = f"https://quotes.toscrape.com/page/{page}/"
26     response = requests.get(url)
27
28     soup = BeautifulSoup(response.text, "html.parser")
29
30     quotes = soup.find_all("span", class_ = "text")
31     authors = soup.find_all("small", class_ = "author")
32
33     for quote, author in zip(quotes, authors):
34         if quote is not None:
35             quote_text = quote.text
36         else:
37             quote_text = "Not Found"
38
39         if author is not None:
40             author_text = author.text
41         else:
42             author_text = "Not Found"
43
44         data.append(
45             {
46                 "Quote": quote_text,
47                 "Author": author_text
48             }
49         )
50         print("Quote:", quote_text)
51         print("Author:", author_text)
52         print()
53
54 df = pd.DataFrame(data)
55 df.drop_duplicates(inplace = True)
56 df.to_csv("Quotes.csv", index = False)
```

```
Release Notes: 1.127.0  posts.csv X
posts.csv > data
1  userId,id,title,body
2  1,1,sunt aut facere repellat provident occaecati excepturi optio reprehenderit,"quia et suscipit
3  suscipit recusandae consequuntur expedita et cum
4  reprehenderit molestiae ut ut quas totam
5  nostrum rerum est autem sunt rem eveniet architecto"
6  1,2,qui est esse,"est rerum tempore vitae
7  sequi sint nihil reprehenderit dolor beatae ea dolores neque
8  fugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis
9  qui aperiam non debitis possimus qui neque nisi nulla"
10 1,3,ea molestias quasi exercitationem repellat qui ipsa sit aut,"et iusto sed quo iure
11 voluptatem occaecati omnis eligendi aut ad
12 voluptatem doloribus vel accusantium quis pariatur
13 molestiae porro eius odio et labore et velit aut"
14 1,4,eum et est occaecati,"ullam et saepe reiciendis voluptatem adipisci
15 sit amet autem assumenda provident rerum culpa
16 quis hic commodi nesciunt rem tenetur doloremque ipsam iure
17 quis sunt voluptatem rerum illo velit"
18 1,5,nesciunt quas odio,"repudiandae veniam quaerat sunt sed
19 alias aut fugiat sit autem sed est
20 voluptatem omnis possimus esse voluptatibus quis
21 est aut tenetur dolor neque"
22 1,6,dolorem eum magni eos aperiam quia,"ut aspernatur corporis harum nihil quis provident sequi
23 mollitia nobis aliquid molestiae
24 perspiciatis et ea nemo ab reprehenderit accusantium quas
25 voluptate dolores velit et doloremque molestiae"
26 1,7,magnam facilis autem,"dolore placeat quibusdam ea quo vitae
27 magni quis enim qui quis quo nemo aut saepe
28 quidem repellat excepturi ut quia
29 sunt ut sequi eos ea sed quas"
30 1,8,dolorem dolore est ipsam,"dignissimos aperiam dolorem qui eum
```

```
Quotes.csv X
Quotes.csv > data
1  Quote,Author
2  "The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking.",Albert Einstein
3  ""It is our choices, Harry, that show what we truly are, far more than our abilities.""J.K. Rowling
4  "There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle.",Albert Einstein
5  ""The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid.""Jane Austen
6  ""Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring.""Marilyn Monroe
7  "Try not to become a man of success. Rather become a man of value.",Albert Einstein
8  "It is better to be hated for what you are than to be loved for what you are not.",André Gide
9  "I have not failed. I've just found 10,000 ways that won't work.""Thomas A. Edison
10 "A woman is like a tea bag; you never know how strong it is until it's in hot water.",Eleanor Roosevelt
11 ""A day without sunshine is like, you know, night.""Steve Martin
12 ""This life is what you make it. No matter what, you're going to mess up sometimes, it's a universal truth. But the good part is you get to decide how you're going to mess up.""J.K. Rowling
13 ""It takes a great deal of bravery to stand up to our enemies, but just as much to stand up to our friends.""J.K. Rowling
14 ""If you can't explain it to a six year old, you don't understand it yourself.""Albert Einstein
15 ""You may not be her first, her last, or her only. She loved before she may love again. But if she loves you now, what else matters? She's not perfect."
16 "I like nonsense, it wakes up the brain cells. Fantasy is a necessary ingredient in living.""Dr. Seuss
17 ""I may not have gone where I intended to go, but I think I have ended up where I needed to be.""Douglas Adams
18 ""The opposite of love is not hate, it's indifference. The opposite of art is not ugliness, it's indifference. The opposite of faith is not here
19 ""It is not a lack of love, but a lack of friendship that makes unhappy marriages.""Friedrich Nietzsche
20 ""Good friends, good books, and a sleepy conscience: this is the ideal life.""Mark Twain
```